🔷 EXPERIMENT 2 — FULL GIT WORKFLOW

🎯 Task Summary

- Even/Odd program (main branch)
- Reverse string program (GIT-EXAM branch)
- Compare branches
- Create Pull Request (PR)

✅ STEP 0: Open Git Bash

```
cd ~
```

✅ STEP 1: Create project folder

```
mkdir data-science-project2
cd data-science-project2
```

✅ STEP 2: Initialize Git repository

```
git init
```

✅ STEP 3: Create program1.py (Even/Odd)

```
touch program1.py
notepad program1.py
```

✍️ Code (Even or Odd):

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

✅ STEP 4: Commit first program

```
git add .
git commit -m "Even Odd program added"
```

✅ STEP 5: Connect GitHub repository

```
git remote add origin https://github.com/your-username/data_science_project2.git
git remote -v
```

✅ STEP 6: Push to GitHub (main/master)

```
git push -u origin main
```

⚠️ If branch is master:

```
git push -u origin master
```

✅ STEP 7: Create new branch (GIT-EXAM)

```
git checkout -b GIT-EXAM
git branch
```

✅ STEP 8: Create program2.py (Reverse String)

```
touch program2.py
notepad program2.py
```

✍️ Code (Reverse String):

```
text = input("Enter a string: ")
reversed_text = text[::-1]
print("Reversed string:", reversed_text)
```

✅ STEP 9: Commit branch changes

```
git add .
git commit -m "Reverse string program added"
```

✅ STEP 10: Push branch to GitHub

git push -u origin GIT-EXAM

✅ STEP 11: Switch back to main/master

git checkout main

(or)

git checkout master

✅ STEP 12: Compare branches

git diff main GIT-EXAM

If you are using master:

git diff master GIT-EXAM

📌 What diff shows:

- program2.py exists only in GIT-EXAM
- main/master only has program1.py

✅ STEP 13: View commit history

git log --oneline --graph --all

🔷 STEP 14: Create Pull Request (IMPORTANT STEP)

📌 On GitHub Create new repo data-science-project2

1. Go to your repository:

https://github.com/your-username/data-science-project2

2. You will see:

"Compare & pull request" button

3. Click it
4. Set:
    - Base branch: main (or master)
    - Compare branch: GIT-EXAM
5. Click:

Create Pull Request

6. Then click:

Merge Pull Request

7. Confirm merge

🔷 FINAL RESULT

🟢 main/master branch:

program1.py → Even/Odd program

🔵 GIT-EXAM branch:

program1.py
program2.py → Reverse string program

After merge:

main/master will also contain program2.py